

Лабораторная работа № 8

Туннелирование IPv6 через IPv4



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Дисциплина: Сетевые технологии

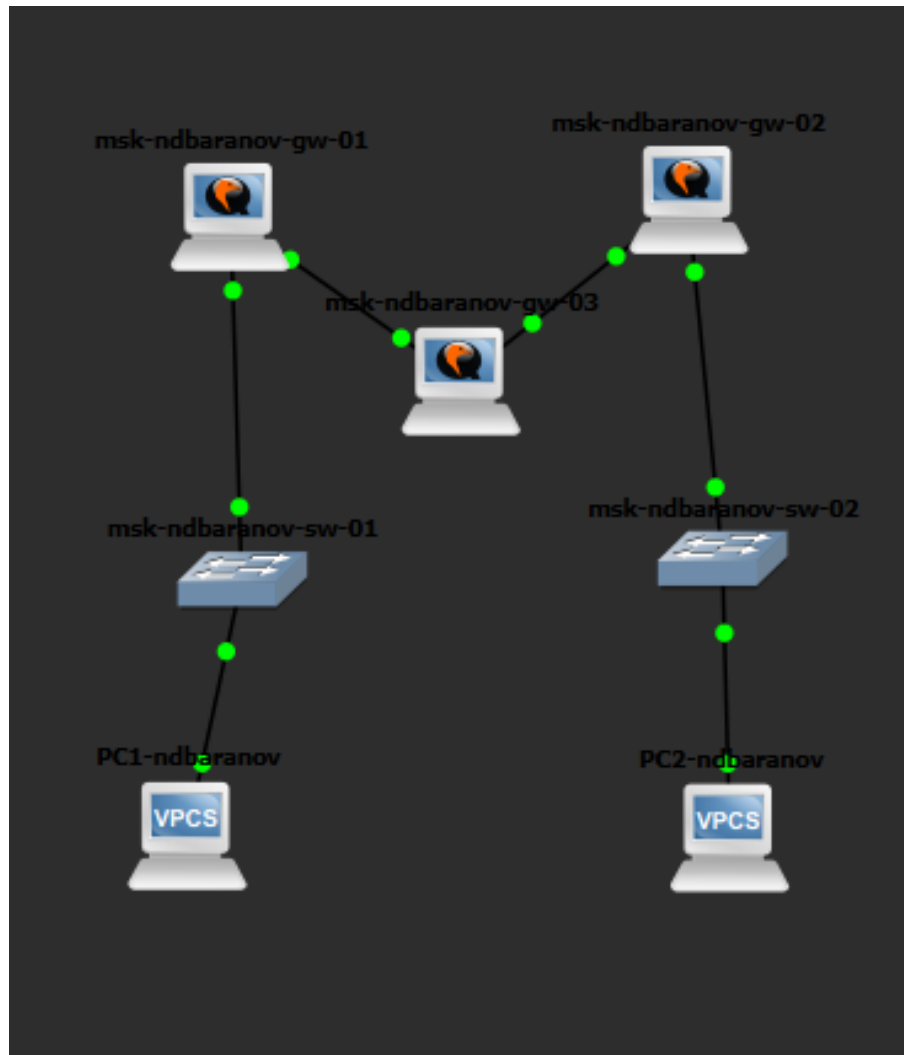
Цель

Настроить туннель между IPv6-сегментами через IPv4-инфраструктуру и исследовать инкапсуляцию.

Основные этапы

Собрана топология из двух IPv6-сетей и промежуточного IPv4-канала. Wireshark подтверждает инкапсуляцию пакета IPv6 внутри IPv4.

Практический результат 1



Полученный результат

IPv6-острова связаны через промежуточную IPv4-сеть.

Практический результат 2

```
PC1-ndbaranov - PuTTY
Executing the startup file
Hostname is too long. (Maximum 12 characters)
VPCS> ip 1000::a/64
PC1 : 1000::a/64
VPCS> save
Saving startup configuration to startup.vpc
. done
VPCS> show ipv6
NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 30030
RHOST:PORT          : 127.0.0.1:30031
MTU                 : 1500
VPCS>

PC2-ndbaranov - PuTTY
Executing the startup file
Hostname is too long. (Maximum 12 characters)
VPCS> ip 1002::a/64
PC1 : 1002::a/64
VPCS> save
Saving startup configuration to startup.vpc
. done
VPCS> show ipv6
NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 30032
RHOST:PORT          : 127.0.0.1:30033
MTU                 : 1500
VPCS>
```

Полученный результат

Крайние VPCS адресованы префиксами 1000::/64 и 2000::/64.

Практический результат 3

The image displays three terminal windows, each representing a VyOS gateway configuration. The windows are titled 'msk-ndbaranov-gw-02 - PuTTY', 'msk-ndbaranov-gw-01 - PuTTY', and 'msk-ndbaranov-gw-03 - PuTTY'. Each window shows the same sequence of commands: 'configure', 'set interfaces ethernet eth0 address', 'set service router-advert interface eth0 prefix', 'set interfaces ethernet eth1 address', 'commit', 'save', and 'exit'. The configurations are for three different IP ranges: gw-01 uses 1000::1/64 and 10.0.0.1/8; gw-02 uses 1002::1/64 and 20.0.0.1/8; gw-03 uses 10.0.0.2/8 and 20.0.0.1/8. A fourth terminal window in the foreground shows the 'Welcome to VyOS!' message and the 'configure' command, with a network diagram visible in the background.

```
msk-ndbaranov-gw-02 - PuTTY
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-ndbaranov-gw-02:~$ configure
[edit]
vyos@msk-ndbaranov-gw-02# set interfaces ethernet eth0 address 1002::1/64
[edit]
vyos@msk-ndbaranov-gw-02# set service router-advert interface eth0 prefix 1002::1/64
[edit]
vyos@msk-ndbaranov-gw-02# set interfaces ethernet eth1 address 20.0.0.1/8
[edit]
vyos@msk-ndbaranov-gw-02# commit
sa[edit]
vyos@msk-ndbaranov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ndbaranov-gw-02# exit
exit
vyos@msk-ndbaranov-gw-02:~$

msk-ndbaranov-gw-01 - PuTTY
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-ndbaranov-gw-01:~$ configure
[edit]
vyos@msk-ndbaranov-gw-01# set interfaces ethernet eth0 address 1000::1/64
[edit]
vyos@msk-ndbaranov-gw-01# set service router-advert interface eth0 prefix 1000::1/64
[edit]
vyos@msk-ndbaranov-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
[edit]
vyos@msk-ndbaranov-gw-01# commit
save[edit]
vyos@msk-ndbaranov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ndbaranov-gw-01# exit
exit

msk-ndbaranov-gw-03 - PuTTY
Welcome to VyOS!
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-ndbaranov-gw-03:~$ configure
[edit]
vyos@msk-ndbaranov-gw-03# set interfaces ethernet eth0 address 10.0.0.2/8
[edit]
vyos@msk-ndbaranov-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8
[edit]
vyos@msk-ndbaranov-gw-03# commit
sav[edit]
vyos@msk-ndbaranov-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ndbaranov-gw-03# exit
```

Полученный результат

Граничные VyOS получили пользовательские и транзитные адреса.

Практический результат 4

```
msk-ndbaranov-gw-01 - PuTTY
edit]
yos@msk-ndbaranov-gw-01# set service router-advert interface eth0 prefix 1000::
64
edit]
yos@msk-ndbaranov-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
edit]
yos@msk-ndbaranov-gw-01# commit
ave[edit]
yos@msk-ndbaranov-gw-01# save
aving configuration to '/config/config.boot'...
one
edit]
yos@msk-ndbaranov-gw-01# exit
xit
yos@msk-ndbaranov-gw-01:~$ trace 10.0.0.2
raceroute to 10.0.0.2 (10.0.0.2), 30 hops max, 60 byte packets
1 10.0.0.2 (10.0.0.2) 2.568 ms 2.126 ms 2.788 ms
yos@msk-ndbaranov-gw-01:~$ trace 20.0.0.1
raceroute to 20.0.0.1 (20.0.0.1), 30 hops max, 60 byte packets
onnect: Network is unreachable
yos@msk-ndbaranov-gw-01:~$ trace 20.0.0.2
raceroute to 20.0.0.2 (20.0.0.2), 30 hops max, 60 byte packets
onnect: Network is unreachable
yos@msk-ndbaranov-gw-01:~$
```

Полученный результат

До создания туннеля доступен только IPv4-транспорт.

Практический результат 5

```
vyos@msk-ndbaranov-gw-01:~$ trace 10.0.0.2
traceroute to 10.0.0.2 (10.0.0.2), 30 hops max, 60 byte packet
 1  10.0.0.2 (10.0.0.2)  1.171 ms  1.505 ms  1.407 ms
vyos@msk-ndbaranov-gw-01:~$ trace 20.0.0.1
traceroute to 20.0.0.1 (20.0.0.1), 30 hops max, 60 byte packet
 1  20.0.0.1 (20.0.0.1)  1.163 ms  0.837 ms  0.919 ms
vyos@msk-ndbaranov-gw-01:~$ trace 20.0.0.2
traceroute to 20.0.0.2 (20.0.0.2), 30 hops max, 60 byte packet
 1  10.0.0.2 (10.0.0.2)  0.941 ms  1.011 ms  0.931 ms
```

Полученный результат

Один IPv4-переход соединяет локальную и удалённую точки туннеля.

Практический результат 6

```
msk-ndbaranov-gw-01 - PuTTY
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-ndbaranov-gw-01:~$ ping 1001::2
PING 1001::2(1001::2) 56 data bytes
64 bytes from 1001::2: icmp_seq=1 ttl=64 time=2.02 ms
64 bytes from 1001::2: icmp_seq=2 ttl=64 time=2.11 ms
64 bytes from 1001::2: icmp_seq=3 ttl=64 time=2.12 ms
64 bytes from 1001::2: icmp_seq=4 ttl=64 time=2.17 ms
64 bytes from 1001::2: icmp_seq=5 ttl=64 time=2.05 ms
64 bytes from 1001::2: icmp_seq=6 ttl=64 time=2.35 ms
64 bytes from 1001::2: icmp_seq=7 ttl=64 time=2.11 ms
64 bytes from 1001::2: icmp_seq=8 ttl=64 time=2.12 ms
64 bytes from 1001::2: icmp_seq=9 ttl=64 time=2.09 ms
64 bytes from 1001::2: icmp_seq=10 ttl=64 time=2.18 ms
64 bytes from 1001::2: icmp_seq=11 ttl=64 time=2.15 ms
64 bytes from 1001::2: icmp_seq=12 ttl=64 time=2.11 ms
64 bytes from 1001::2: icmp_seq=13 ttl=64 time=2.20 ms
64 bytes from 1001::2: icmp_seq=14 ttl=64 time=2.06 ms

msk-ndbaranov-gw-02 - PuTTY
vyos@msk-ndbaranov-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-ndbaranov-gw-02# set interfaces tunnel tun0 address 1001::2/64
[edit]
vyos@msk-ndbaranov-gw-02# commit
sav[edit]
vyos@msk-ndbaranov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ndbaranov-gw-02# exit
exit
vyos@msk-ndbaranov-gw-02:~$ ping 1001::1
PING 1001::1(1001::1) 56 data bytes
64 bytes from 1001::1: icmp_seq=1 ttl=64 time=2.08 ms
64 bytes from 1001::1: icmp_seq=2 ttl=64 time=2.05 ms
64 bytes from 1001::1: icmp_seq=3 ttl=64 time=2.14 ms
64 bytes from 1001::1: icmp_seq=4 ttl=64 time=2.27 ms
64 bytes from 1001::1: icmp_seq=5 ttl=64 time=1.98 ms
64 bytes from 1001::1: icmp_seq=6 ttl=64 time=2.04 ms
64 bytes from 1001::1: icmp_seq=7 ttl=64 time=2.18 ms
64 bytes from 1001::1: icmp_seq=8 ttl=64 time=2.14 ms
64 bytes from 1001::1: icmp_seq=9 ttl=64 time=2.13 ms
```

Полученный результат

Маршруты к чужим IPv6-префиксам
направлены в tunnel-интерфейс.

Практический результат 7

```
mस्क-ndbaranov-gw-01 - PuTTY
[edit]
vyos@msk-ndbaranov-gw-01# commit
sa[edit]
vyos@msk-ndbaranov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ndbaranov-gw-01# exit
exit
vyos@msk-ndbaranov-gw-01:~$ show ipv6 route
Codes: K - kernel route, C - connected, S - static, R - RIPng,
       O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
       v - VNC, V - VNC-Direct, A - Babel, D - SHARP, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
C>* 1000::/64 is directly connected, eth0, 00:07:07
C>* 1001::/64 is directly connected, tun0, 00:01:55
S>* 1002::/64 [1/0] via 1001::2, tun0, weight 1, 00:00:22
C * fe80::/64 is directly connected, tun0, 00:01:55
C * fe80::/64 is directly connected, eth0, 00:10:55
C * fe80::/64 is directly connected, eth1, 00:10:56
C>* fe80::/64 is directly connected, lo, 00:10:58
vyos@msk-ndbaranov-gw-01:~$

mस्क-ndbaranov-gw-02 - PuTTY
[edit]
vyos@msk-ndbaranov-gw-02# commit
sav[edit]
vyos@msk-ndbaranov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ndbaranov-gw-02# exit
exit
vyos@msk-ndbaranov-gw-02:~$ show ipv6 route
Codes: K - kernel route, C - connected, S - static, R - RIPng,
       O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
       v - VNC, V - VNC-Direct, A - Babel, D - SHARP, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
S>* 1000::/64 [1/0] via 1001::1, tun0, weight 1, 00:00:20
C>* 1001::/64 is directly connected, tun0, 00:01:50
C>* 1002::/64 is directly connected, eth0, 00:06:47
C * fe80::/64 is directly connected, tun0, 00:01:50
C * fe80::/64 is directly connected, eth0, 00:08:05
C * fe80::/64 is directly connected, eth1, 00:08:06
C>* fe80::/64 is directly connected, lo, 00:08:07
vyos@msk-ndbaranov-gw-02:~$
```

Полученный результат

После настройки IPv6 проходит между удалёнными сегментами.

Практический результат 8

Захват с Standard input [msk-ndbaranov-gw-01 Ethernet1 to msk-ndbaranov-gw-03 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.0.2	224.0.0.9	RIPv2	66	Response
2	23.003939	10.0.0.2	224.0.0.9	RIPv2	66	Response
3	42.408671	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0c00, seq=1/256, ttl=64 (reply in 4)
4	42.409395	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0c00, seq=1/256, ttl=64 (request in 3)
5	43.408183	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0c00, seq=2/512, ttl=64 (reply in 6)
6	43.408890	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0c00, seq=2/512, ttl=64 (request in 5)
7	44.410260	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0c00, seq=3/768, ttl=64 (reply in 8)
8	44.410943	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0c00, seq=3/768, ttl=64 (request in 7)
9	45.410328	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0c00, seq=4/1024, ttl=64 (reply in 10)
10	45.411040	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0c00, seq=4/1024, ttl=64 (request in 9)
11	46.006056	10.0.0.2	224.0.0.9	RIPv2	66	Response
12	46.411895	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0c00, seq=5/1280, ttl=64 (reply in 13)
13	46.412515	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0c00, seq=5/1280, ttl=64 (request in 12)
14	47.413772	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0c00, seq=6/1536, ttl=64 (reply in 15)
15	47.414600	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0c00, seq=6/1536, ttl=64 (request in 14)
16	47.736717	0c:6d:e8:17:00:01	0c:20:01:3a:00:00	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
17	47.737536	0c:20:01:3a:00:00	0c:6d:e8:17:00:01	ARP	60	10.0.0.2 is at 0c:20:01:3a:00:00
18	47.743408	0c:20:01:3a:00:00	0c:6d:e8:17:00:01	ARP	60	Who has 10.0.0.1? Tell 10.0.0.2

Полученный результат

Wireshark показывает исходный IPv6-пакет внутри IPv4.

Итог

Настроить туннель между IPv6-сегментами через IPv4-инфраструктуру и исследовать инкапсуляцию. Полученные результаты подтверждают выполнение поставленной задачи.